

A New Variable Neighborhood Search Algorithm for the Multi Depot Heterogeneous Vehicle Routing Problem with Time Windows

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Abstract

The Multi Depot Heterogeneous Vehicle Routing Problem with Time Windows (MDHVRPTW) is an important variant of the Vehicle Routing Problem (VRP), which is complex and still not resolved well. To solve the problem, mathematical model of the MDHVRPTW is constructed and a modified variable neighborhood search algorithm is proposed. In the algorithm, a hybrid operator of insert and exchange are used to achieve the shaking process, the Best-improvement strategy is adopted, and it can make the algorithm to achieve a better balance in the solution quality and running time. The performance of the proposed algorithm is tested on the benchmark instances proposed by Golden and compared with other algorithms in the literature. The results indicate that the proposed algorithm is competitive and feasible in solving the MDHVRPTW.

Keywords: Multi depot; Heterogeneous; Vehicle routing problem; Variable neighborhood search

1 Introduction

The multi depot heterogeneous vehicle routing problem with time windows (MDHVRPTW) [2] is a variant of the vehicle routing problem (VRP) where the vehicles do not necessary have the same capacity and they belong to different distribution centers or depots. Therefore, the MDHVRPTW involves designing a set of vehicle routes, each starting and ending at the depot, for a heterogeneous fleet of vehicles which services a set of customers with known demands. The remainder of the paper is organized as follows: the problem definition and mathematical formulation are discussed in Section 2. Section 3 introduces the main ideas of the new variable neighborhood search. Computational results are presented and discussed in Section 4. Section 5 concludes the paper with a resume of the approach.

2 Problem descriptions

2.1 Problem definition

The number of customers is denoted by n and the number of depots is denoted by m . Thus, the problem is defined on a complete graph $G = (V, E)$, where $V = \{v_1, \dots, v_n, v_{n+1}, \dots, v_{m+n}\}$ is the vertex set and $E = \{(v_i, v_j) : v_i, v_j \in V, i \neq j\}$ is the arc set. The customer set $C = \{v_1, \dots, v_n\}$ represents customers, while vertices set $D = \{v_{n+1}, \dots, v_{m+n}\}$ corresponds to m depots. Each vertex $v_i \in V$ has several non-negative weights associated with it, namely, a demand q_i , a service time s_i , as well as an earliest e_i and latest l_i possible start time for the service, which define a time window $[e_i, l_i]$. Furthermore, the depot vertices v_{n+1} to v_{m+n} feature no demands and service times, i.e. $q_i = s_i = 0; \forall i \in 1, \dots, m$. Associated to C_{ijk}^l is the transportation cost from customer i to customer j for k vehicles of l type. The delivery vehicles set $T = \{T_1^1, \dots, T_1^k, \dots, T_{LK}^1\}$ corresponds to the set of k vehicles for l type. Each l type vehicle k has associated a non-negative capacity D_k^l and non-negative maximum route duration S_k^l . w_k^l is the capacity of k vehicles for l type. d_{ij} is the linear distance from customer i to customer j . R_m is the capacity of m depots. ρ_i is penalty cost for unit-time violations of the specified time window for customer node i , ρ_k is penalty cost for unit-time violations of the maximum working time for vehicle k , Δa_i represents i th-time

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window violation due to early service, Δb_i ith-time window violation due to late service, ΔB_k corresponds to working time violation for vehicle k . A_i is vehicle arrival time at node i .

2.2 Problem mathematical formulation

The problem objective (1) aims to minimize the overall service expenses, including traveling distance and time costs, waiting and service time costs and penalty costs.

$$\min \sum_{i \in D \cup C} \sum_{j \in D \cup C} \sum_{k \in T} C_{ijk}^l X_{ijk}^l d_{ij} + \rho_k \Delta B_k + \sum_{i \in C} \rho_i (\Delta a_i + \Delta b_i) \quad (1)$$

s.t.

$$\sum_{l, k \in T} \sum_{i \in D \cup C} X_{ijk}^l = l, j \in C; \quad (2)$$

$$\sum_{i \in C} \sum_{j \in D \cup C} q_i X_{ijk}^l \leq w_k^l, l, k \in T \quad (3)$$

$$\sum_{i \in C} q_i Z_{ij} - \sum_{m \in D} R_m \leq 0 \quad (4)$$

$$\sum_{i \in D \cup C} X_{ijk}^l = Y_{ik}^l, j \in D \cup C, l, k \in T \quad (5)$$

$$\sum_{j \in D \cup C} X_{ijk}^l = Y_{jk}^l, i \in D \cup C, l, k \in T \quad (6)$$

$$\sum_{m \in D} \sum_{i \in C} X_{ijk}^l \leq 1, l, k \in T \quad (7)$$

$$S_{ik} - S_{jk} + nX_{ijk}^l \leq n - 1, i, j \in C, l, k \in T \quad (8)$$

$$\sum_{l, k \in T} \sum_{i \in D \cup C} X_{ijk}^l d_{ij} \leq D_k^l, j \in C \quad (9)$$

$$\Delta a_i \geq e_i - A_i, \forall i \in C \quad (10)$$

$$\Delta b_i \geq A_i - l_i, \forall i \in C \quad (11)$$

3 A new variable neighborhood search algorithm

3.1 Initial solution

Assume h_{ij} represents the distance from distribution center i to the customer j , the distance set is $H_{ij} = \{h_{ij} | i = 1, 2, \dots, m; j = 1, 2, \dots, n\}$. The minimum value is $\min H_{ij}$ in the distance set, the second minimum is $\text{second} \min H_{ij}$. Sum_k denotes the total number of customers that vehicles k delivers the goods, the set $U_k = \{y_{ik} | i = 0, 1, \dots, \text{Sum}_k\}$ corresponds to the vehicle k to server the customers, y_{ik} presents the vehicle k as a transportation for customer i , y_{0k} means the distribution center (depot) is the initial point of the vehicle k . Steps of the initial feasible solution are as follows:

Step 1, calculate $r_j = \min H_{ij} / \text{second} \min H_{ij}$, $1 \leq i \leq m$, and select the appropriate ε ($0 \leq \varepsilon \leq 0$) ;

Step 2, if $r_j < \varepsilon$, customer j will assign to the distribution center (depot) corresponding to $\min H_{ij}$;

Step 3, repeat Step 1 and Step 2, traverses all customers point to form m customer groups;

Step 4, randomly take a customer group, the vehicle initial residual loading is $W_k^l = w_k^l$, $k = 0$, Sum_k , $U_k = \emptyset$;

Step 5, the demand of the customer i is q_i , and $k = 1$;

Step 6, if $w_k^l \leq q_i$, $W_k^l = q_i - w_k^l$; else goto step 9;

Step 7, if $S_{i-1} + S_i \leq S_k$, $U_k = U_k \cup \{i\}$, $\text{Sum}_k = \text{Sum}_k + 1$; otherwise goto Step 9;

Step 8, if $k = K$, $k = K$; else $k = k$;

Step 9, $k = k + 1$, goto step 6;

Step 10, $i = i + 1$, goto Step 5;

Step 11, repeat Step 5 to Step 10, K records the total of being used vehicle, R_k corresponds to the feasible route.

3.2 Shaking

In the shaking execution, it first selects a neighborhood structure N_k from the set of neighborhood structures of current solution x , then according to the definition of N_k , x corresponds to change and generate a new solution x^* . There are two neighborhood structures to achieve the shaking: insert and exchange. The insert and exchange operators are shown in Figure 1.

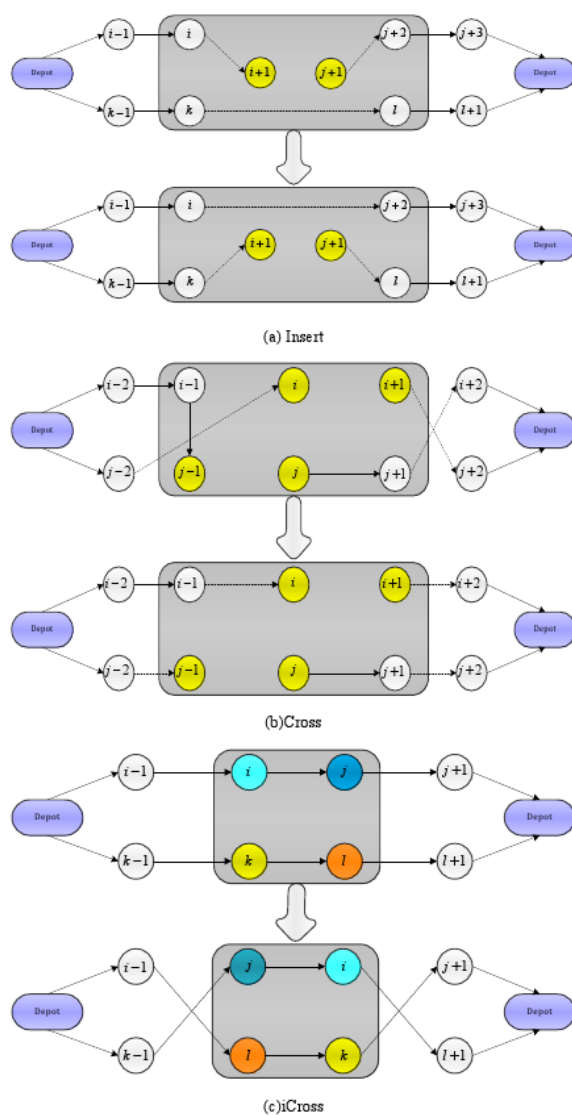


Fig. 1. Insert and exchange operator

In each neighborhood the insert operator is applied with a probability p_{insert} to both routes to further increase the extent of the perturbation, then the probability of the exchange operator is $1 - p_{insert}$. IVNS selects randomly an exchange operator to change path for each shaking execution.

3.3 Local search

In VNS algorithm [4,3], local search procedure will search the neighborhood of the new solution space obtained through shaking in order to achieve a locally optimal solution. Based on the previous studies, this paper selects $2-opt$ and $Or-opt$ as a local search operator in order to obtain the good quality local optimal solution in a short period. According to the probability, one of the two operators is selected in each local search process. The parameter p_{2-opt} represents the probability of selection for $2-opt$, similarly, the probability of selection for $Or-opt$ can be expressed as $1 - p_{2-opt}$. In this paper, we adopt the best-improvement strategy, it enables the algorithm to achieve a better balance in the solution quality and run time.

3.4 Acceptance decision

To avoid that the VNS becomes too easily trapped in local optima, this is accomplished by utilizing a Metropolis criterion like in simulated annealing for inferior solutions x^* and accepts them with a probability of Eq. (15). We update T every n_T iterations by an amount of $T_{n+1} = \delta T_n$, where q_0 a random number on the interval $[0,1]$, Where δ is settable cooling coefficient, and an initial temperature value is $T_0 = 10$.

$$SA(x^*, x) = \begin{cases} x^*, & \text{if } q_0 > \exp\left(\frac{f(x^*) - f(x)}{T}\right) \\ x, & \text{if } q_0 \leq \exp\left(\frac{f(x^*) - f(x)}{T}\right) \end{cases} \quad (12)$$

4 Numerical experiments

4.1 Problem data and experimental setting

IIVNS algorithm is implemented by the C # language, and the main configuration of the computer is Intel Core i3 1.8GHz, 2 GB RAM and Window XP. The benchmark problem is used to test the performance of the algorithm, and it is 12 of 20 issues proposed by Golden et al. The initial values of the various parameters for IVNS algorithm are set as follows:

(1) the parameter settings for simulated annealing accepted criteria are initial temperature $T_0 = 10$, every $n_T = n/10$ generation to update temperature $T_{n+1} = 0.9 \times T_n$, $n_t = 1000$ to end the algorithm.

(2) the parameters value of the Shaking operation are as follows: $p_{insert} = 0.2$, $p_{cross} = 0.15$, $p_{icross} = 0.1$.

(3) the p_{2-opt} value is 0.5 in local search.

4.2 Numerical results

The results in Table 1 show that the proposed algorithms can get nine best solutions, ARPD is 0.12. Grendreau with their algorithms have found six known best solution of the problem. The ARPD of Grendreau [6] is 0.03. Brandao [1], Imran [5] find the nine and eight best solutions of the questions respectively, ARPD are 0.13 and 0.72, slightly higher than our algorithm.

Table 1
Comparison of solution quality from the different methods

No.	n	Best	Grendreau (1999)		Brandao (2008)		Imran, et al. (2009)		IVNS	
			Cost	CPU	Cost	CPU	Cost	CPU	Cost	CPU
G3	20	961.03	961.03	164	961.03	21	961.03	21	961.03	0
G4	20	6437.33	6445.10	253	6437.33	22	6437.33	18	6437.33	1
G5	20	1007.05	1007.05	164	1007.05	20	1007.05	13	1007.05	2
G6	20	6516.47	6516.47	309	6516.47	25	6516.47	22	6516.47	2
G13	50	2406.36	2406.36	724	2406.36	145	2406.36	252	2406.36	21
G14	50	9119.03	9125.65	1033	9119.03	220	9119.03	274	9119.03	19
G15	50	2586.37	2606.72	901	2586.37	110	2586.37	303	2586.72	23
G16	50	2720.43	2720.43	815	2720.43	111	2741.50	253	2720.43	20
G17	75	1734.53	1734.53	1022	1744.83	322	1745.33	745	1743.76	52
G18	75	2369.65	2412.56	691	2371.49	267	2369.65	897	2369.65	50
G19	100	8659.74	8685.71	1687	8664.29	438	8665.12	1613	8664.81	175
G20	100	4039.49	4166.73	1421	4039.49	601	4066.94	1595	4039.49	108
Average		4046.46	4065.70	765.33	4047.85	191.83	4051.85	500.50	4047.68	39.42
BestNumd ¹			6		9		8		9	
ARPDd ²			0.03		0.72		0.13		0.12	

¹ ARPD-average relative percentage deviation

² BestNum-the number of best solution

5 Conclusions

We presented a Variable Neighborhood Search (VNS) metaheuristic for the Multi Depot Heterogeneous Vehicle Routing Problem with Time Windows (MDHVRPTW) a generalized variant of the classical VRP. The performance of our method is shown when comparing our best solution values to the benchmark instances proposed by Golden. The results indicate that the proposed algorithm is effective in solving the MDHVRPTW. Finally, this study shows that our method can be applied successfully in the large water project problems.

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